

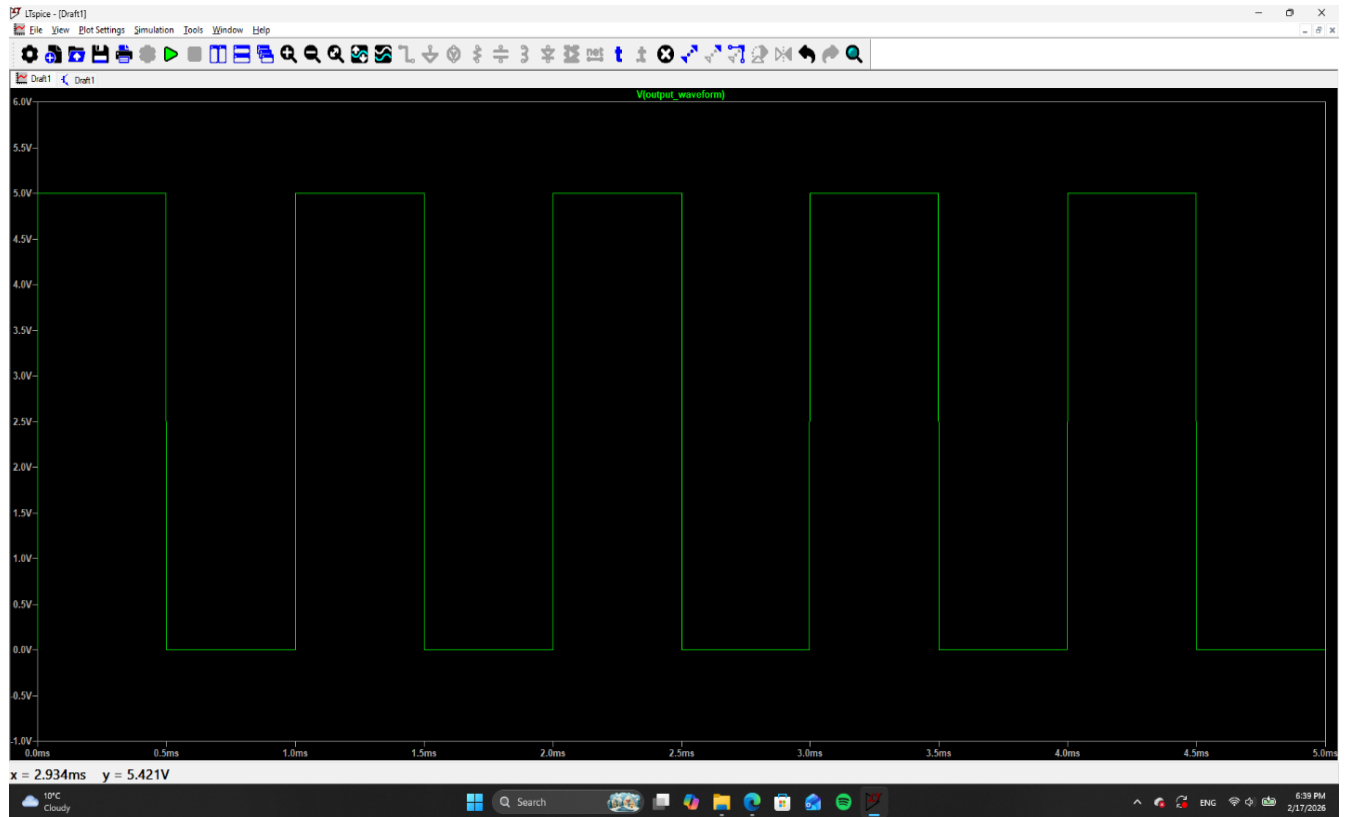
RLC Time Domain Analysis

ENGS152 Circuits Lab

21 February 2026

Generated Square Wave

Below you can see the plot of the output waveform created by the square wave. To prove that it has a frequency of 1kHz we can look at the graph and see that it takes 1ms for the graph to complete 1 full cycle. Hence 1ms is the period of the generated square wave. We can apply the $f = \frac{1}{T}$ to further prove it. $f = \frac{1}{0.001\text{ s}} = 1000\text{ Hz} = 1\text{ kHz}$. By knowing that the frequency is 1kHz we can also find the period without looking at the graph, by just applying the formula backwards and finding it. $T = \frac{1}{1000\text{ Hz}} = 1\text{ ms}$.



Capacitor Charge/Discharge

Calculations of the R and C values

For this part of the pre-lab, we are asked to find R and C such the time constant $\tau=0.1\text{ms}$.

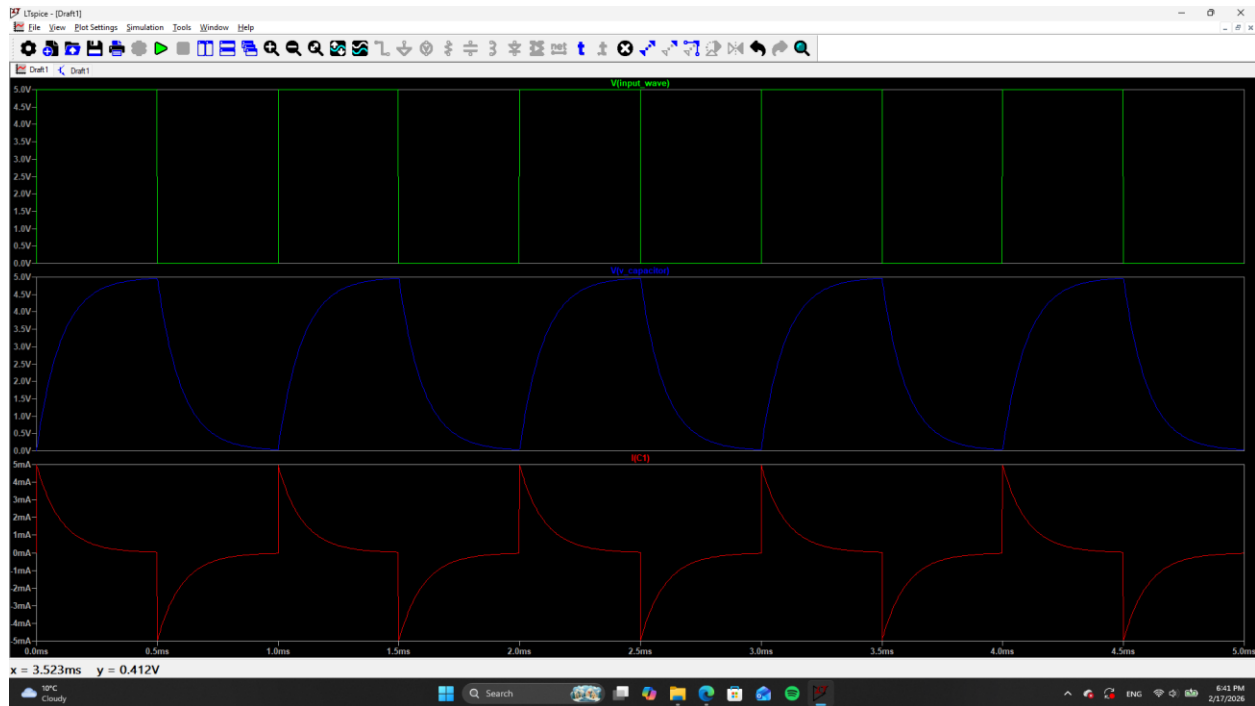
$$\tau = RC = 0.1 \text{ ms}$$

We can start by choosing a resistor value. $1\text{k}\Omega$ is a nice practical common in lab value to start from. We applied the formula and solved it for C.

$$C = \frac{\tau}{R} = \frac{1 \times 10^{-4}}{1000} = 100 \text{ nF}$$

Such that, our selected R and C values satisfy the required time constant.

Plot evaluation



As the wave goes from Low to High, the capacitor begins to charge, and the voltage increases exponentially and reaches the input voltage with a time constant of 0.1 ms. The current starts decreasing exponentially towards zero as the capacitor becomes charged and two power sources oppose each other.

As the input wave voltage goes from High to Low, the capacitor discharges and the voltage decrease exponentially towards zero. However, the current reverses the direction and increases exponentially towards zero from the negative axis.

Inductor Charge/Discharge

Calculations of the R and L values

For this part of the pre-lab, we are asked to find R and L such the time constant $\tau=0.1\text{ms}$.

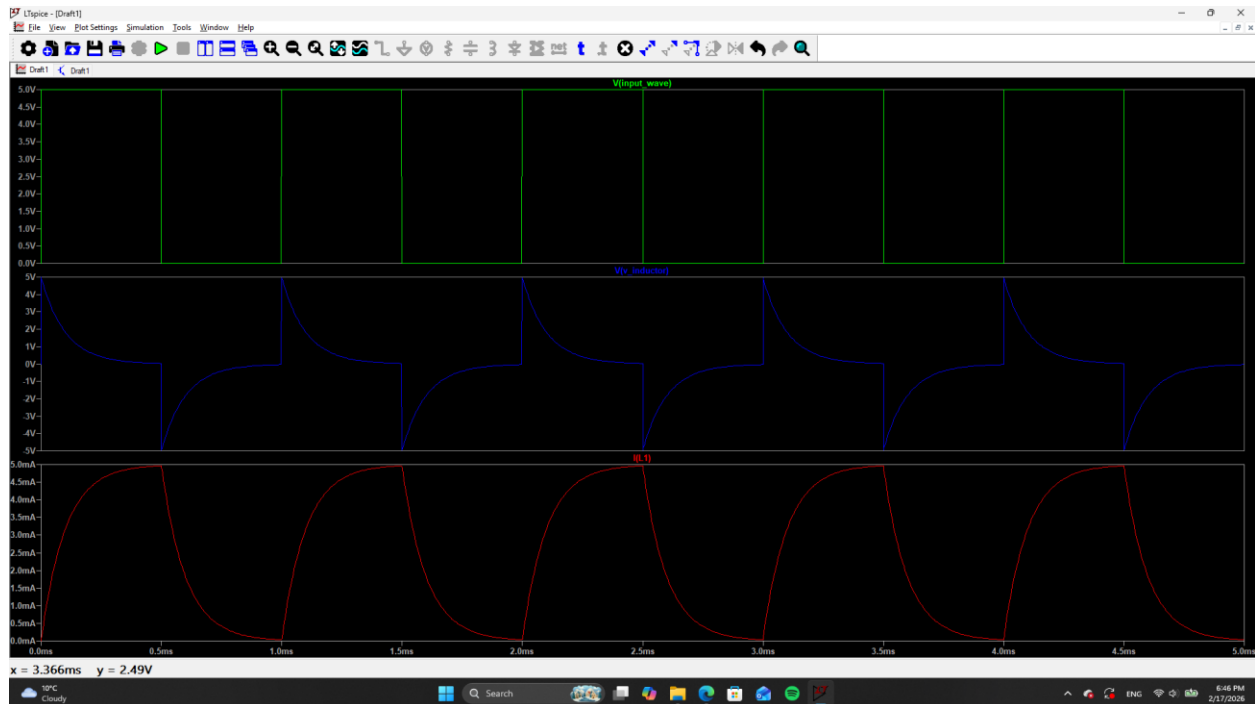
$$\tau = \frac{L}{R} = 0.1 \text{ ms}$$

We can start by choosing a resistor value. $1\text{k}\Omega$ is a nice practical common, in lab value to start from. We applied the formula and solved it for C.

$$L = \tau R = 0.0001 \times 1000 = 0.1 \text{ H}$$

Such that, our selected L and R values satisfy the required time constant.

Plot evaluation



As the input switches from Low to High, the inductor starts opposing the change in the circuit. The current starts increasing exponentially to a steady state with a time constant of 0.1 ms, and then

the voltage across the inductor decreases to zero as that steady state is reached. It is important to understand that the voltage in the inductor depends on the rate of change of current in the inductor, not how big the current is.

As the input voltage switches from High to Low, the current does not drop instantly and starts decreasing towards zero exponentially. The voltage reverses its polarity and goes towards zero.